

Electric charge and field.

electrostatic electricity → Study charge at Rest.

- **Thales** → When Amber rod rubbed with woollen clothes. So Rod attract dust particle. firstly observed by him.
- **William Gilbert** → attract and repels object. charge apply force and experience.
- **Thomas Edison** → Gave term electricity.
- electrified body or Charged body Contain charge.
- on electrification a body is said to have accept charge.
- **Benjamin Franklin** → term positive/negative
- Ebonite rod [-ve charge] woollen cloth [+ve charge]
Glass rod [+ve charge] Silk cloth [-ve charge]

DATE _____
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Charge

property of matter due to which it experience force

- SI unit **Coulomb [C]**
- Scalar Quantity.
- Dimensional formula **AT**
- Same charge Repel and opposite attract each other.
- Unit charge / Basic charge / fundamental charge
Elementary charge $1.6 \times 10^{-19} \text{ C}$.
- Polarity of charge $\rightarrow$ Differential between positive and negative charge.
- When electron Remove from atom it acquire positive charge (mass decrease)
- When atom gain electron it acquire negative charge (mass increase)
- Charge is invariant Does not depend on velocity.

Tyres of aeroplane get charged at time of landing due to friction b/w earth and tyres.
So tyres made Slightly Conducting, So Charge transfer to earth.

properties of electric Charge.

(i) Additivity of charge

total electric charge on an object is equal to algebraic sum of all electric charge

$$Q = Q_1 + Q_2 + Q_3 \dots$$

Also known as Principle of Superposition of Charge.

(ii) Charge is conserved

total charge of Isolated system is always conserved.

it is not possible to create or destroy net charge of system.

E.x → pair production and pair annihilation.
 γ -split ← γ -ray → $e^- + e^+$ [Positron] $e^- + e^+ \rightarrow \gamma + \gamma$

Emit γ -ray Photon

(iii) Quantisation of Charge

total charge on body is always integral multiple of Basic unit charge

$$m = n m_e \quad Q = ne \quad Q = ne \quad n = 1, 2, 3, \dots$$

$$\rightarrow 9.1 \times 10^{-31} \text{ kg.}$$

- In one Coulomb of charge there is 6×10^{18} electron.

No. of

(iv) Charge is always associated with mass

Conductor $\rightarrow$ Have free electron
Substance through which electric current
can flow easily.
Ex $\rightarrow$ metal, human body, etc.

Silver is best conductor of electricity.

Insulator $\rightarrow$ Do not have free electron
Substance through which electric
charge cannot flow.

Ex $\rightarrow$ Rubber, plastic etc
free electron are not found in insulator.

Charged particle.
At Rest $\rightarrow$ only electric field
At motion $\rightarrow$ electric field and
magnetic field
• $V = \text{constant}$
• $V \neq \text{constant}$ $\rightarrow$ electric field, magnetic field
accelerated charge and electromagnetic wave
with Radiat energy.

Charge distribute uniformly on surface
Dielectric constant is maximum for air. and always greater than 1.
dielectric $\rightarrow$ are non-conductor.
Do not have free e^-

If medium change then force reduce by
 $\frac{F}{E}$

Coulomb's law.

"force between two point charge is directly proportional to product of charge and inversely proportional to square distance between them"

$$F \propto q_1 q_2$$

$$F \propto \frac{1}{r^2}$$

$$F \propto \frac{q_1 q_2}{r^2}$$

$K \rightarrow$ proportionally constant
 $9 \times 10^9 \text{ Nm}^2/\text{C}^2$

$$F = \frac{K q_1 q_2}{r^2}$$

$$K = \frac{1}{4\pi \epsilon_0}$$

Depend on medium, unit.

$$F = \frac{1}{4\pi \epsilon_0} \frac{q_1 q_2}{r^2}$$

$\epsilon_0 \rightarrow$ Permittivity of free space

$$8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

$$\boxed{\epsilon = \epsilon_0 \epsilon_r} \rightarrow \begin{array}{l} \text{Relative permittivity} \\ \text{or Dielectric constant} \end{array}$$

medium

free space

metal $\rightarrow \infty$

Air $\rightarrow 1$

always greater than 1

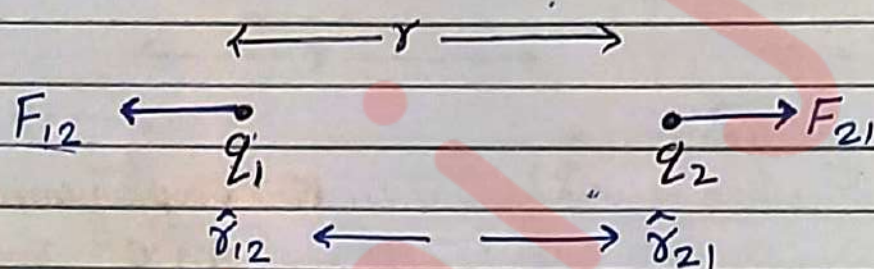
$$\frac{K_1}{K_2} = \frac{F_2}{F_1}$$

1 Coulomb $\rightarrow$ If two like charge of equal magni. placed 1m apart in vacuum and experience force of Repulsion $9 \times 10^9 \text{ N}$.

$$F = 9 \times 10^9 \times (1 \times 1)$$

Coulomb's law in vector form

Considered two point charge q_1 and q_2 placed in vacuum at distance r from each other repel each other.



In vector form

force on charge q_1 due to q_2 .

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12} \quad \text{--- (i)}$$

$\hat{r}_{12} = \frac{r_{12}}{r}$ is unit vector in direction from q_1 to q_2

force on charge q_2 due to q_1

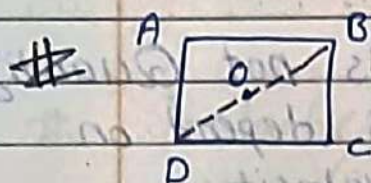
$$F_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{21} \quad \text{--- (ii)}$$

$\hat{r}_{21} = \frac{r_{21}}{r}$ is unit vector in direction from q_2 to q_1

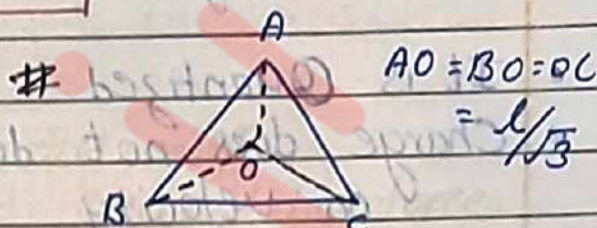
$$\hat{r}_{12} = -\hat{r}_{21}$$

from eq. (i) and (ii)

$$F_{12} = -F_{21}$$



$$DO = OB =$$

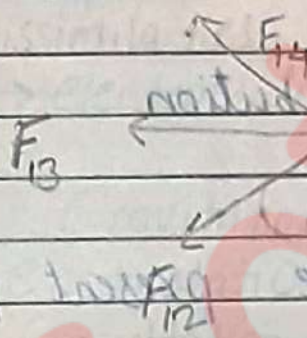


$$AO = BO = CO = \frac{l}{\sqrt{3}}$$

force b/w multiple charge

Superposition principle

total force on given charge due to number of charge is vector sum of individual force acting on that charge



$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14}$$

$$\vec{F}_1 = \frac{q_1 q_2}{4\pi\epsilon_0} \hat{r}_{12} + \frac{q_1 q_3}{4\pi\epsilon_0} \hat{r}_{13} + \frac{q_1 q_4}{4\pi\epsilon_0} \hat{r}_{14}$$

Individual force unaffected due to presence of other charge.

Charge

mass.

- It is +ve, -ve or zero
 - It is always conserved
 - It is Quantized
 - Charge does not depend on velocity
 - force b/w charge may be attractive or repulsive
- It is always +ve
 - It is not conserved.
 - It is not Quantized
 - mass depend on velocity.
 - force b/w mass is attractive.

Continuous Charge Distribution

① linear Charge Distribution (λ)

length

$$\lambda = \frac{Q}{L}$$

Charge present per unit

SI unit $\rightarrow C/m$

② Surface charge Distribution (σ)

unit area



$$\sigma = \frac{Q}{A}$$

Charge present per

SI unit $\rightarrow C/m^2$

③ Volume Charge Distribution (ρ)

Charge present per unit Volume.

$$\rho = \frac{q}{V}$$

SI unit C/m^3

Similarities and Dissimilarities b/w

Gravitational and electrostatic force.

• Similarities

Both obey inverse square law.

Conservative in nature.

Both are central forces.

• Dissimilarities

→ electrostatic force may be attractive or

Repulsive $F = \frac{kq_1q_2}{r^2}$

Gravitational force always attractive

→ electrostatic force is strong in nature

Gravitational force is weak in nature.

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2}$$

$$F_g = \frac{Gm_1m_2}{r^2}$$

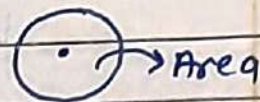
$$F_e = 10^{39} F_g$$

Limitation of Coulomb's law.

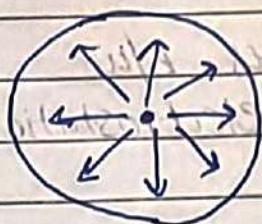
- only applicable charge at rest
- only applicable to point charge

Electric field.

Area around the charge in which its electric effect experienced.



electric field line
lines used to represent electric effect field.

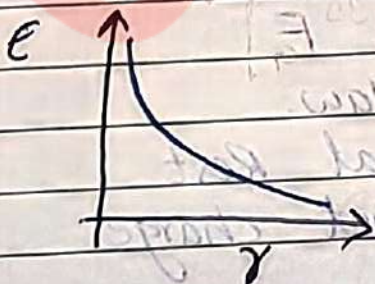


michael farady.

electric field intensity. E
force acting per unit positive small test charge.

$$\vec{E} = \frac{\vec{F}}{q_0} \text{ or } E = \frac{kq}{r^2}$$

- SI unit N/C
- Dimension formula $MLT^{-3}A^{-1}$
- It is vector quantity.

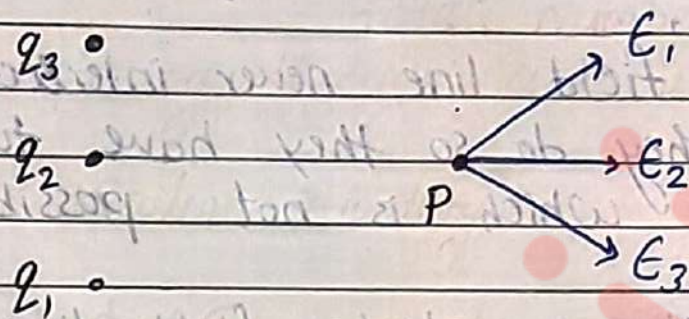


electric field intensity depend on q

electric field Inten. depend on source charge.

$$E_m = \frac{E}{k}$$

Electric field due to system of point charge.

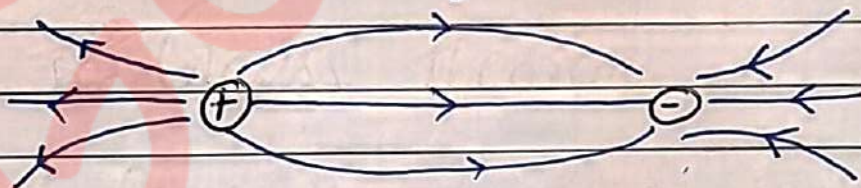


$$E_T = E_1 + E_2 + E_3$$

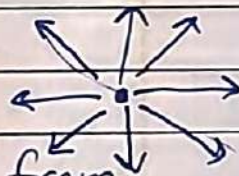
$$= \frac{kq_1}{r_{1p}} + \frac{kq_2}{r_{2p}} + \frac{kq_3}{r_{3p}}$$

properties of electric field line

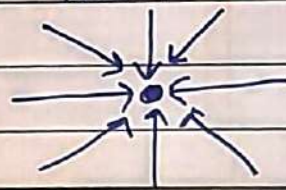
- electric lines start from positive charge and end into negative charge



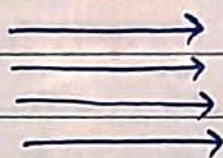
- In case of single positive charge electric lines start from charge and end ~~in~~ into infinite

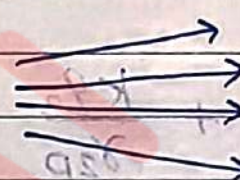


- In case of single negative charge electric line start from infinite and end into charge

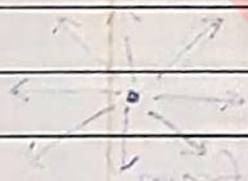
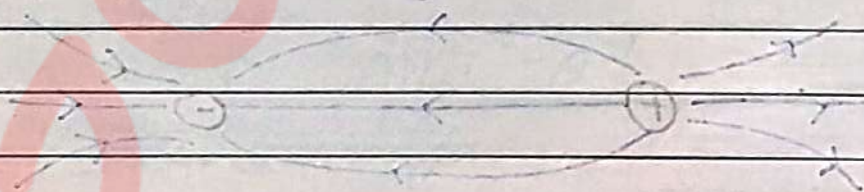


- Tangent drawn at any point gives direction of electric field intensity.
- Two electric field lines never intersect each other if they do so they have two directions at a point which is not possible.
- Near the charge electric field lines are dense and intensity is strong vice versa.

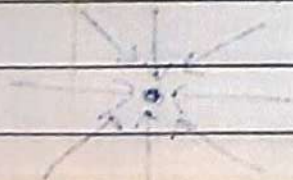

 uniform
 electric field


 non uniform
 electric field.

Electric lines start from positive charge and end into negative charge.



In case of single positive charge electric lines start from it and go into infinity.



In case of single negative charge electric lines start from infinity and go into it.

Electric flux Φ

total number of electric field line passing through area.

- It is scalar quantity.

$$\Phi \propto E$$

$$\Phi \propto A.$$

$$\Phi = EA \cos \theta$$

$$\Phi_{\max} \rightarrow \theta = 0^\circ / 180^\circ$$

$$\Phi_{\min} \rightarrow \theta = 90^\circ$$

$$\Phi = \int E \cdot ds.$$

- SI unit Nm^2/C or $V \cdot m$.

- Dimension formula $ML^3T^{-3}A^{-1}$

- If θ less than $90^\circ \rightarrow \Phi \rightarrow +ve$

- —||— more —||— $\rightarrow \Phi \rightarrow -ve$

Gauss Theorem

total flux through closed surface is $\frac{1}{\epsilon_0}$ times the net charge inside closed surface

$$\Phi = \frac{Q_{\text{net}}}{\epsilon_0}$$

always Remember about direction

$$90^\circ \rightarrow 0$$

$$0^\circ \rightarrow \text{max}$$

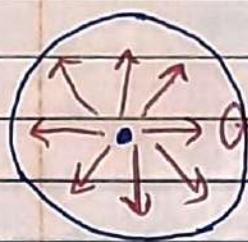
→ flux passing through cube each face

$$\phi = \frac{q_{in}}{6\epsilon_0}$$

→ electric flux when Dipole inclosed in surface is Zero.

Proof of Gauss theorem.

electric flux through a surface. is



$$\phi = \int E \cdot ds \cos \theta$$

$$= \int E \cdot ds$$

$$\theta = 0$$

$$\cos \theta = 1$$

$$\Rightarrow \frac{kq}{r^2} \int ds \quad E = \frac{kq}{r^2}$$

$$\Rightarrow \frac{kq}{r^2} \int_0^{4\pi r^2} ds$$

$$1 ds = x$$

$$\Rightarrow \frac{kq}{r^2} [x]_0^{4\pi r^2}$$

$$K = \frac{1}{4\pi\epsilon_0}$$

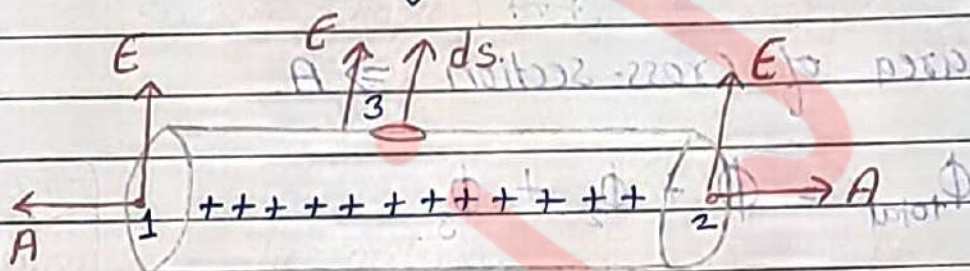
$$\Rightarrow \frac{1 \times q \times 4\pi r^2}{4\pi\epsilon_0 r^2}$$

$$\phi = \frac{q}{\epsilon_0}$$

Application of Gauss theorem

① field due to infinite long straight charged wire

Considered an infinite long thin straight wire with linear charge density (λ)



from Gauss law $\phi = \int E ds = \frac{q}{\epsilon_0}$
Now,

$$\phi_{\text{total}} = \phi_1 + \phi_2 + \phi_3$$

$$= \int EA \cos 90^\circ + \int EA \cos 90^\circ + \int EA \cos 0^\circ$$

$$A \cdot 0 = 0 \Rightarrow \int_0^{2\pi r l} E ds$$

$$\boxed{\phi_E \Rightarrow E 2\pi r l}$$

$$\boxed{\cos 90^\circ = 0}$$

Charge enclosed in cylinder. $\lambda = \frac{q}{l}$ $q = \lambda l$

$$\phi = \frac{q}{\epsilon_0}$$

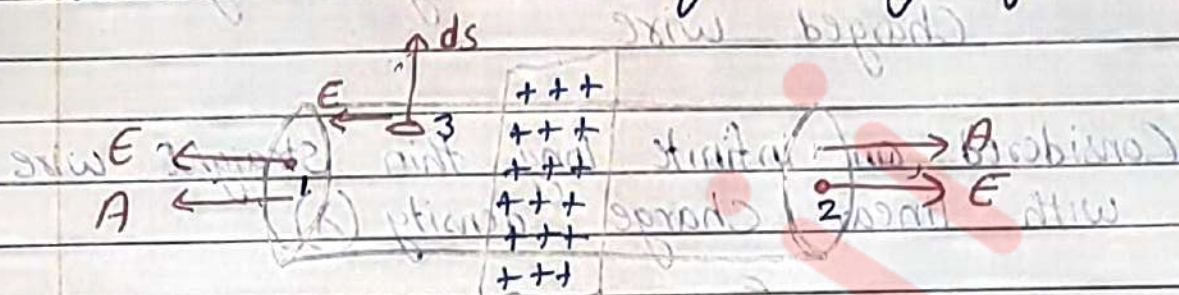
$$E 2\pi r l = \frac{q}{\epsilon_0}$$

$$E 2\pi r l = \frac{\lambda l}{\epsilon_0}$$

$$\boxed{E = \frac{\lambda}{2\pi \epsilon_0 r}}$$

* field due to thin infinite plan sheet charge

let σ be surface charge density of sheet.



area of cross-section $\Rightarrow A$.

$$\Phi_{\text{total}} = \Phi_1 + \Phi_2 + \Phi_3$$

$$= EA \cos \theta + EA \cos \theta + EA \cos \theta$$

$$= EA \cos 0 + EA \cos 0 + EA \cos 90$$

$$\Rightarrow EA + EA + 0$$

$$\boxed{\Phi_{\text{total}} = 2EA}$$

Gauss theorem $\Phi = \frac{Q}{\epsilon_0}$

Surface charge density $\sigma = \frac{Q}{A} \Rightarrow Q = \sigma A$

$$\Phi_{\text{total}} \Rightarrow \frac{\sigma A}{\epsilon_0}$$

$$2EA = \frac{\sigma A}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

Electric field intensity due to uniform charged thin spherical shell.

Consider a thin metallic spherical shell of radius r and charge distribute on surface

Hence

$$\text{Surface charge } \sigma = \frac{q}{A} \Rightarrow \sigma = \frac{q}{4\pi r^2} \quad (i)$$

Case I.

when it lie outside $r < R$.



$$\phi = \int E ds \cos \theta$$

$$= \int_0^{4\pi R^2} E ds$$

$$\Rightarrow E \int ds$$

$$\phi \Rightarrow E 4\pi R^2 \quad (ii)$$

Now

according to Gauss law

$$\phi = \frac{q}{\epsilon_0}$$

$$\phi = \frac{\sigma 4\pi r^2}{\epsilon_0} \quad (iii)$$

$$\left[q = \sigma 4\pi r^2 \right] \text{ about } \epsilon q \quad (i)$$

from eq. (ii) (iii)

$$E 4\pi R^2 = \frac{\sigma 4\pi r^2}{\epsilon_0}$$

$$E = \frac{\sigma r^2}{\epsilon_0 R^2}$$

Case II on the Surface

$$r = R$$

$$E = \frac{\sigma r^2}{\epsilon_0 R}$$

$$E = \frac{\sigma}{\epsilon_0}$$

Case III

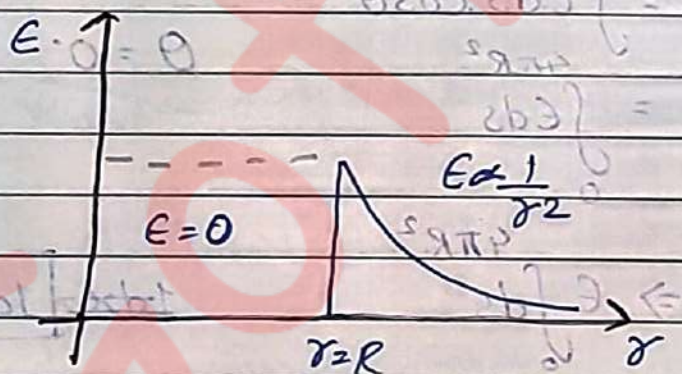
Inside the Surface

$$r < R$$

$$E = 0$$

$$\phi = 0$$

because No + Charge present inside Surface.



$$\phi = \int E \cdot dr$$

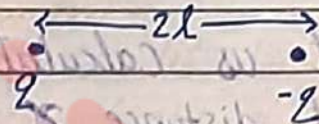
$$\phi = \int E \cdot dr$$

$$\phi = \int E \cdot dr$$

$$\phi = \int E \cdot dr$$

Electric Dipole

A pair of equal and opposite point charge, are separated by short distance



Electric Dipole moment (P)

Product of magnitude of one charge and distance between charge:

$$P = q \cdot 2l$$

- It is vector quantity. - from (-ve to +ve)
- SI unit is C-m.
- Dimension formula ATL.
- Size of Ideal dipole is Zero

$$\rightarrow 2q = 0, q = \infty \text{ Hence}$$

Dipole field $\rightarrow$

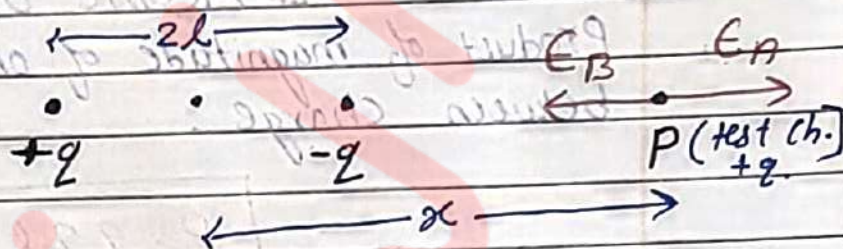
Electric field produce by electric dipole.

$\rightarrow$ Practical unit of dipole moment is debye (D)

$\rightarrow$ Net charge on dipole is Zero but intensity is not may Zero.

Electric field at axial point of electric Dipole.

let us calculate electric field at point P.
at distance x from centre of dipole.



$$E_p = E_B - E_A$$

$$E = kq \frac{1}{(x-l)^2} - kq \frac{1}{(x+l)^2}$$

$$\Rightarrow kq \left[\frac{1}{(x-l)^2} - \frac{1}{(x+l)^2} \right]$$

$$\Rightarrow kq \left[\frac{(x+l)^2 - (x-l)^2}{(x+l)^2(x-l)^2} \right]$$

$$\Rightarrow \frac{kq 4xl}{(x^2 - l^2)^2}$$

$$\Rightarrow \frac{kq 4xl}{(x^2)^2}$$

Short dipole $x \gg l$

$$= \frac{kq 2l 2x}{x^4}$$

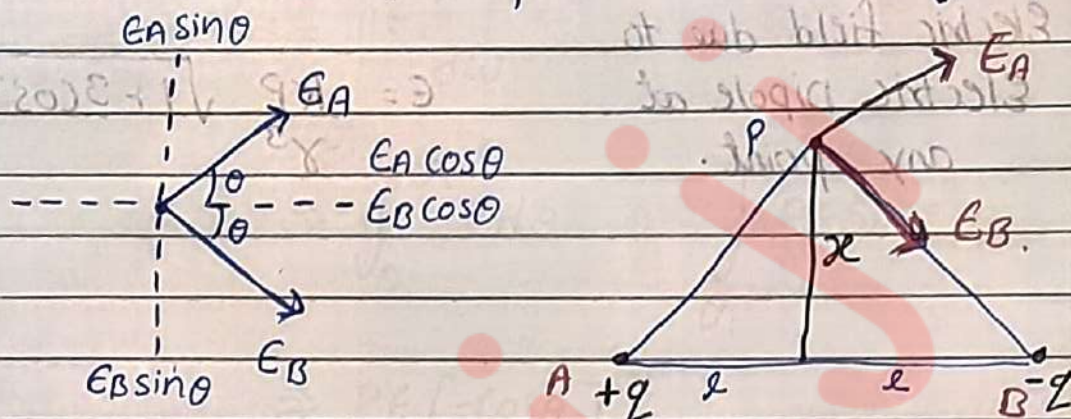
$$\Rightarrow \frac{2kP}{x^3}$$

$$P = q 2l$$

$$E = \frac{2kP}{x^3}$$

Electric field due to electric Dipole at Equatorial line.

Considered we have electric Dipole, let take a point P at distance x apart from centre of dipole



→ value of vertical component

then electric field is Zero

→ Now net electric field is only due to electric Horizontal Component.

$$E_N = E_A \cos \theta + E_B \cos \theta$$

$$= \frac{2Kq}{r^2} \times \frac{l}{\sqrt{x^2 + l^2}}$$

$$\Rightarrow \frac{2Kql}{(x^2 + l^2)(\sqrt{x^2 + l^2})}$$

$$\Rightarrow \frac{2Kp}{(x^2 + l^2)^{3/2}}$$

$$\Rightarrow \frac{Kp}{x^3}$$

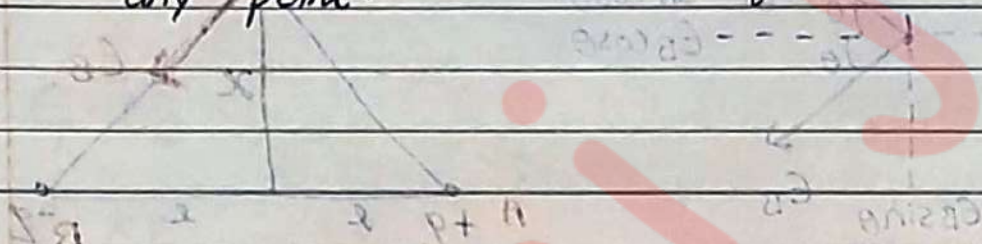
$$x \gg l$$

$$E = \frac{Kp}{x^3}$$

$$\frac{E_{\text{axial}}}{E_{\text{equatorial}}} = \frac{2}{1}$$

Electric field due to electric dipole at any point.

$$E = \frac{KP}{r^3} \sqrt{1 + 3\cos^2\theta}$$



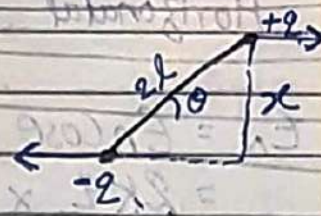
force on Dipole.

$$F = qE$$

Net force on dipole is Zero in uniform electric field.

Torque on electric dipole in uniform electric field.

$T = \text{magnitude of force} \times \text{perpendicular distance}$



$$T \Rightarrow qE 2l \sin\theta$$

$$\sin\theta = \frac{x}{2l}$$

$$T \Rightarrow E \times P \sin\theta$$

$$x = 2l \sin\theta$$

N-m

$$\vec{\tau} = \vec{E} \times \vec{P}$$

Direction Right hand Thumb Rule.

Case I

$$\theta = 0^\circ \quad T = 0$$

$\vec{P}$, $\vec{E}$ are parallel

Dipole stable

Equilibrium.

Case II

$$\theta = 90^\circ \quad T = \text{max.}$$

Dipole suffer

maximum Torque.

Case III

$$\theta = 180^\circ \quad T = 0$$

$\vec{P}$, $\vec{E}$ antiparallel

Dipole

unstable Equilib.

Work done in Rotating an Electric Dipole in a Uniform Electric field.

→ vector quantity.

$$dw = \tau d\theta$$

$$= \int_{\theta_1}^{\theta_2} dw$$

$$\Rightarrow \int_{\theta_1}^{\theta_2} \tau d\theta \Rightarrow \int_{\theta_1}^{\theta_2} PE \sin\theta d\theta$$

$$\Rightarrow PE [-\cos\theta]_{\theta_1}^{\theta_2}$$

$$\Rightarrow PE [-\cos\theta_2 - (-\cos\theta_1)]$$

$$\Rightarrow PE [\cos\theta_1 - \cos\theta_2]$$

$$W = PE [\cos\theta_1 - \cos\theta_2]$$

potential energy of an electric Dipole in uniform electric field.

$$U = -PE \cos\theta$$

Case-I

$$\theta = 0^\circ$$

U = minimum.

$$U = -PE$$

Stable Equilibrium

Case-II

$$\theta = 90^\circ$$

U = Zero

U = Zero

Standard position

Case III

$$\theta = 180^\circ$$

U = maximum

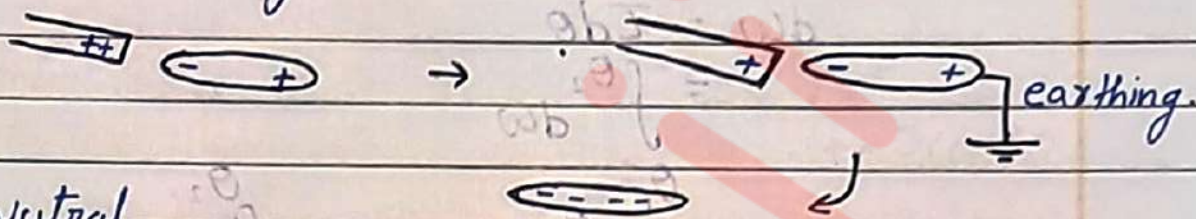
U = PE

unstable Equilibrium.

method of charging

(i) Induction $\rightarrow$

When a charged Conductor brought near to an uncharged Conductor.

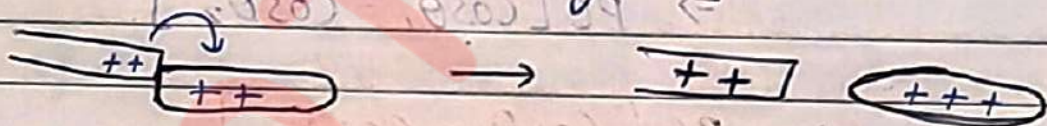


~~Neutral~~ Body Get charged without actual touch and gets opposite charge.

(ii)

Conduction $\rightarrow$

When a charged conductor brought near and touch uncharged Conductor.



Neutral Body Gets same charged with actual touch and get same charge.

(iii) friction $\rightarrow$ Body get charge by ~~At~~ Rubbing opposite in Nature

Gold leaf electroscope.

Device used to detect the presence of charge

